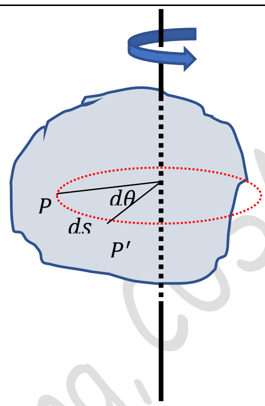
ROTATION OF RIGID BODIES

Kinematics of rotation

When a body rotates about a fixed axis, different mass elements of the body will be having different displacements, velocity and accelerations. But each mass element will be moving along circular paths centered about the axis of rotation.

Angular Displacement

The change in the angular position of any point P on the body defined by $d\theta$ is called angular displacement. When a rigid body rotates, every point in the body will have the same angular displacement. If the point P is at a distance r from the axis of rotation, the displacement of that point $ds = r\,d\theta$. Now, we can understand that different points at different distances from the axis will have different displacements, though they have the same angular displacement $d\theta$



Angular Velocity

The time rate of change in the angular position is called the angular velocity .

$$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} = \dot{\theta}$$

But , we know $ds = r\,d\theta$ ⋙⋙ $\frac{ds}{r} = d\theta$

$$\omega = \frac{d\theta}{dt} = \frac{1}{r}\frac{ds}{dt} = \frac{v}{r}$$

Angular acceleration

The time rate of change of angular velocity is called the angular acceleration

$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} = \ddot{\theta}$$

As discussed in the curvilinear translation, the resultant acceleration of a mass element dm can be resolved in to tangential and normal (to the circular path it describes) components.

$$a_n = \frac{v^2}{r} = r\omega^2 = r\dot{\theta}^2$$

$$a_t = \frac{dv}{dt} = \frac{d(r\omega)}{dt} = r\frac{d\omega}{dt} = r\ddot{\theta}$$

When a body rotates with constant angular velocity, its tangential acceleration is zero.

Then its normal acceleration is directed towards the axis of rotation. It is also known as centripetal acceleration.

Suppose the angular acceleration is constant, $\ddot{\theta} = a\ constant$

$$\frac{d\omega}{dt} = \ddot{\theta}$$

$$d\omega = \ddot{\theta}\,dt$$

Integrating both sides knowing that when $t = 0, \omega = \omega_0$

$$\int_{\omega_0}^{\omega} d\omega = \ddot{\theta}\int_0^t dt$$

$$\omega - \omega_0 = \ddot{\theta}t$$

$$\omega = \omega_0 + \ddot{\theta}t$$

$$\omega = \omega_0 + \ddot{\theta}t$$

$$\frac{d\theta}{dt} = \omega_0 + \ddot{\theta}t$$

$$d\theta = (\omega_0 + \ddot{\theta}t)\,dt$$

Integrating both sides knowing that the initial angular position is taken as zero

$$\int_0^{\theta} d\theta = \int_0^t (\omega_0 + \ddot{\theta}t)\,dt$$

$$\theta = \omega_0\,t + \frac{1}{2}\,\ddot{\theta}t^2$$

If the angular velocity is a function of angular displacement $[\omega = f(\theta)]$

$$\ddot{\theta} = \frac{d\omega}{dt} = \frac{d\omega}{d\theta}\frac{d\theta}{dt} = \omega\frac{d\omega}{d\theta}$$

$$\omega\, d\omega = \ddot{\theta}\, d\theta$$

Integrating both sides knowing that the body starts from zero angular position with angular velocity ω_0

$$\int_{\omega_0}^{\omega} \omega\, d\omega = \int_{0}^{\theta} \ddot{\theta}\, d\theta$$

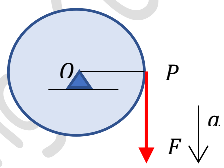
If $\ddot{\theta}$ is constant

$$\frac{\omega^2 - \omega_0^2}{2} = \ddot{\theta}\theta$$

$$\omega^2 = \omega_0^2 + 2\ddot{\theta}\theta$$

Example 1

A cord is wrapped around a wheel of radius $0.2\, m$ initially at rest. If a force is applied to the cord and gives an acceleration $= 4t\ m/s^2$, where t is in seconds, determine the angular velocity and angular position of line OP as function of time.



P is a point on the circumference of the wheel. Any point on the wheel will have a tangential acceleration and a normal acceleration. As the cord is moved tangentially to the drum, the acceleration of the cord is the tangential acceleration of point P.

$$a_t = 4t \quad \ggg\gg \quad r\ddot{\theta} = 4t \quad \ggg \quad \ddot{\theta} = 20\, t$$

The angular acceleration is a function of time

$$\frac{d\omega}{dt} = 20\, t \quad \ggg\ d\omega = 20\, t\, dt$$

Integrating both sides knowing that initially the wheel is at rest

$$\int_0^{\omega} d\omega = \int_0^{t} 20\ t\, dt \ggg\ggg \ \omega = 10\, t^2$$

$$\omega = \frac{d\theta}{dt} = 10\, t^2$$

$$d\theta = 10\, t^2\, dt$$

Integrating both sides assuming that the wheel starts from zero angular position

$$\int_0^{\theta} d\theta = \int_0^{t} 10\, t^2\, dt \quad \ggg \quad \theta = \frac{10}{3}t^3$$

Example 2

A wheel has an initial clockwise angular velocity of $10\, rad/s$ and a constant angular acceleration of $3\, rad/s^2$. Determine the number of revolutions it must undergo to acquire a clockwise angular velocity of $15\, m/s$. What is the time required?

Initial angular velocity, $\omega_0 = 10\, rad/s$

Final angular velocity, $\omega = 15\, rad/s$

Constant angular acceleration, $\ddot{\theta} = 3\, rad/s^2$

Since the angular acceleration is constant, $\omega = \omega_0 + \ddot{\theta}t \quad \ggg\ t = 1.67\, s$

The angular displacement, $\theta = \omega_0 t + \frac{1}{2}\ddot{\theta}t^2 = 10 \times 1.67 + \frac{1}{2} \times 3 \times 1.67^2 = 20.8\, rad$

When the wheel makes one revolution, the angle turned is 2π rad

Number of revolutions made by the wheel, $N = \frac{20.8}{2\pi} = 3.32\, revolutions$

Example 3

The armature of an electric motor has an angular speed $N = 1800\,rpm$ at the instant when the power is cut off. If it comes to rest in $6\,s$, calculate its angular deceleration assuming it to be a constant. How many complete revolutions does the armature make during this period?

Initial angular velocity, $\omega_0 = \frac{2\pi N}{60} = 60\pi\,rad/s$

Final angular velocity, $\omega = 0$

Time $= 6\,s$

angular acceleration, $\ddot{\theta} = ?$

Since $\ddot{\theta}$ is assumed to be a constant, we can use the relation $\omega = \omega_0 + \ddot{\theta}t \ggg \ddot{\theta} = -10\pi \frac{rad}{s^2}$

The angular displacement, $\theta = \omega_0 t + \frac{1}{2}\ddot{\theta}t^2 = 60\pi \times 6 - \frac{1}{2} \times 10\pi \times 6^2 = 180\pi\,rad$

Number of revolutions made by the armature, $N = \frac{180\pi}{2\pi} = 90\,revolutions$

#Exercise 40

40.1

The angle of rotation of a body is given by the equation $\theta = 3t^2 + 4t + 2$. Find the angular velocity and angular acceleration after $2\,s$.

40.2

A rotating wheel requires $3\,s$ to rotate 37 revolutions. Its angular velocity at the end of the $3\,s$ interval is $98\,rad/s$. What is the constant angular acceleration of the wheel?

40.3

A car accelerates uniformly from rest and reaches a speed of 22 m/s in 9 s If the diameter of a tire is 58 cm, find

(a) the number of revolutions the tire makes during this motion, assuming no slipping, and

(b) the final rotational speed of the tire in revolutions per second.